

A dietary risk assessment of Aflatoxin in food based products in Pakistan

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Abstract—Aflatoxins are naturally occurring toxins found/developed in crop-based food products due to development of fungi causing a wide variety of harmful diseases to the consumers of its affected food products. And the most painful part is that very few people are aware of this and mere unawareness regarding aflatoxins is becoming a cause of its widespread. Therefore, as part of this research, we analyzed dietary behavior of a wide range of people from different age, gender, BMI and region groups to determine their aflatoxin consumption and probable health impacts and risks they are having as a result of this intake. We also analyzed the awareness of people regarding aflatoxin consumption and how it is deriving their dietary behaviors. Our analysis shows that the survey participants on average consume $45.736070 \text{ ng/kg}^{-1} \text{ day}^{-1}$ of aflatoxin posing a mean Cancer Risk of 0.921582 cancer/year/100,000 population ng/kg bw/day and mean MOE of 13.142884 which is way below the safe limit of 10,000 as stated by European Food Safety Authority. More unfortunate part is that 78.7% survey respondents had never heard about aflatoxin.

Keywords—aflatoxin, EDI, MOE, Cancer Risk, Pakistan

I. INTRODUCTION

Maize, rice and corn are the most frequently used food products we use either in raw or processed form or as an ingredient in any food product. But rarely we do analyze the concentration of contaminants in these products before consuming them which is becoming the cause of the most harmful disease – liver cancer. Maize, rice and corn are stored and transported for a long period of time until they could reach to the consumers. Poor handling and storage conditions on these points leads to the development of fungi which eventually produces a harmful family of toxin known as aflatoxin that causes wide variety of diseases ranging from growth retardation in children to immune suspension to eventually liver cancer.

Food-based sauces of various types are highly consumed in Pakistan. As a matter of fact, no comprehensive studies have been carried out to assess the dietary risk associated with aflatoxin contamination in foodstuffs that form part of daily diets. Given the generally warm and humid climate of Pakistan, which is conducive for the growth of fungi producing aflatoxins, there is a dire need to evaluate the

health risks posed by these toxins in commonly consumed food items.

The present study was undertaken as an attempt to conduct a dietary risk assessment related to food-based products in Pakistan, focusing particularly on sauces. A full-scale assessment of the levels of aflatoxins in the product and their contributing health risks would provide useful results to food safety regulators and public health officials. These study results would identify some mitigation strategies that could be adopted for effective risk management to ensure the safety and well-being of the population.

II. LITERATURE REVIEW

A. Introduction to Aflatoxins

Aflatoxins are fungi-based contaminants that are produced in very commonly used food products under certain conditions of humidity, temperature and poor storage conditions for the food products. Out of its 20 types identified, four most important fall under B-type and G-type namely AFB1, AFB2 and AFG1, AFG2 respectively. A. flavus is identified to be the source of production for B-types and A. parasiticus is the source of production for G-types[1,2,3].

B. Production of Aflatoxins

Maize is the most widely consumed staple everywhere in the world and especially in regions of Asia and Africa that are affected by climate changes. Besides being the most produced crop, it is the part of everyday meal both in raw and processed form and most commonly in conjunction with other food products or as an ingredient in processed foods. [4] Since it's a crop that has to be harvested, stored and then transported to consumers, it is most susceptible of getting fungi producing aflatoxins due to poor storage conditions and change in climate involved during its harvest to consumption cycle. [5,6,7] Milk is another widely used food product especially most popular as an ingredient in a variety of our favorite food. Milk producing animals most commonly cows, buffaloes and goats are fed on heap and straw that is contaminated with aflatoxin produced by fungi grown in their due to humidity and climate conditions. This aflatoxin infected feed in turn is reflected in the milk produced by these animals. [8] Other than these, peanuts and

rice are also most vulnerable to getting aflatoxin contamination and storage conditions if not regulated properly can heavily contaminate peanuts and rice. [9,10].

C. Health Impacts of Aflatoxins

Aflatoxin is the cause of many harmful diseases. Its variant AFB1 found to be the cause of liver cancer causing 172000 cancer cases every year and in most of the cases it was so sever that it led to the death of patient within months of getting diagnosed with the cancer. [11] Another study conducted on Portugese population revealed mean probable intake of 13.43 ng/kg body weight per day of aflatoxin due through food products mentioned above which in turn give rise to 0.13 extra cases of liver cancer. [12] A study conducted in India showed a correlation between aflatoxin consumption and G to T mutations that in turn gives rise to liver cancer. It also showed its interaction with Hepatitis B and C virus that in turn increases the risk of getting HCC. [13] Other than liver cancer, another most pressing affect of aflatoxin is its impact on children's growth. It is found that higher exposure of aflatoxin to children significantly hampered their height and weight growth with the parameters "Length-for-age Z-score (LAZ)", length, knee-heel length, and "weight-for-age Z-score (WAZ)" all showing negative association with the aflatoxin consumption. In addition, higher levels of aflatoxin increased the risk of stunting in children with the risk increasing by 1.18 times for each unit consumption of AFB1. [14] Therefore, there is a large negative correlation between aflatoxin consumption and growth of children and it severely retards their growth. [15] Additionally, the risk of contamination in milk and milk products, through feed contamination, further amplifies the health risks to consumers, including infants. The urgent need for effective risk assessments is critical to manage aflatoxin exposure and implement better safety standards in Pakistan's food industry. [16]

III. RESEARCH METHODOLOGY

A. Data Collection

For data collection, a questionnaire was created on google form and disseminated to audience varying widely in age, gender, BMI levels and region. In the context of Pakistan, the most commonly used diets containing aflatoxins were determined to be wheat, rice and milk therefore, our questionnaire also focused on determining the intake of food products containing these to better understand and evaluate aflatoxin consumption.

B. Health Risk Characterization

To determine daily consumption of aflatoxin through the diets described above, "Estimated Daily Intake (EDI)" was determined using the formula taken from [10] for each of the diets:

$$EDI = (\text{Daily Intake} \times \text{Mean Concentration}) / \text{Weight}$$

Mean concentration of aflatoxin in wheat and rice have been taken from [17] and milk has been taken from [18]. The values used are as follows:

TABLE I. MEAN AFLATOXIN CONCENTRATION IN SOME FOOD PRODUCTS IN PAKISTAN

Food	Mean Aflatoxin Concentration (µg/kg)
Wheat	14.3533
Rice	12.955
Raw Milk	1.476
Packaged Milk	0.245

Having determined EDI, first risk characterization parameter determined is "Margin of Exposure (MOE)" which basically compares toxin exposure to the allowable limit as determined by a standard and then determines whether the exposure is within safe limits or not. In this case, the standard with which we compared aflatoxin consumption is "Benchmark Dose Lower Confidence Limit (BMDL)" for 10% exposure of aflatoxin. This determines the safe limit of daily exposure of aflatoxin to male rats and is determined to be $0.4\mu\text{gkg}^{-1}\text{day}^{-1}$ by European Food Safety Authority. Formula for calculating MOE as mentioned by [10] is:

$$MOE = BMDL_{10} / EDI$$

Second health risk parameter determined is "Cancer Risk". For determining Cancer Risk, average potency was first found using JECFA formula:

$$\text{Average Potency} = (0.3 \times \text{HBsAg-ve prevalence}) + (0.01 \times \text{HBsAg + ve prevalence})$$

HBsAg + ve prevalence used in above formula is 3.5% as determined by [19] for Pakistani population and thus HBsAg-ve prevalence used is 96.5%. Cancer Risk is calculated as:

$$\text{Cancer risk} = EDI \times \text{Average potency}$$

C. Data Analysis

First of all descriptive statistical analysis has been performed to determine different metrics of data including mean, median, minimum and maximum values for EDI of each food product, MOE and Cancer Risk. This is followed by the evaluation of the impact of aflatoxin consumption in terms of MOE and Cancer Risk on each age, gender, BMI, and region group is studied to determine most impacted user group.

Secondly, correlation analysis is performed to determine which food product among the three under analysis (wheat, rice, milk) has the most strong relation and thus contribution to MOE and Cancer Risk.

Thirdly, hypothesis testing has been performed to study the impact of aflatoxin consumption on each category of categorical variables and to understand whether group means for MOE and Cancer risk are equal or not. For the categorical variable "Gender", T-test is used since it only has two categories for which t-test suffices the requirement. For the other categories including Age, BMI and Region having more than 3 categories, ANOVA test is used. For both the tests:

Null Hypothesis (H_0): All group means are equal

Alternative Hypothesis (H_1): At least one group mean is different.

And null hypothesis is rejected if $p \leq 0.05$

Lastly, we use confidence intervals for estimating population means for EDI, MOE and Cancer Risk.

IV. RESULTS

A. Descriptive Statistics

Figures 1-4 show the proportion of each category of user groups that took part in the survey conducted.

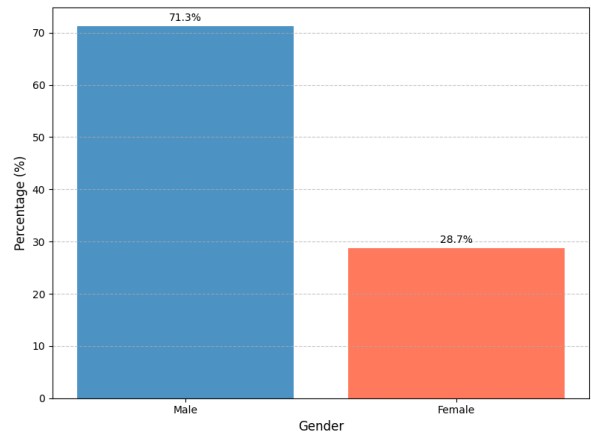


Fig. 1. Gender distribution in survey participants

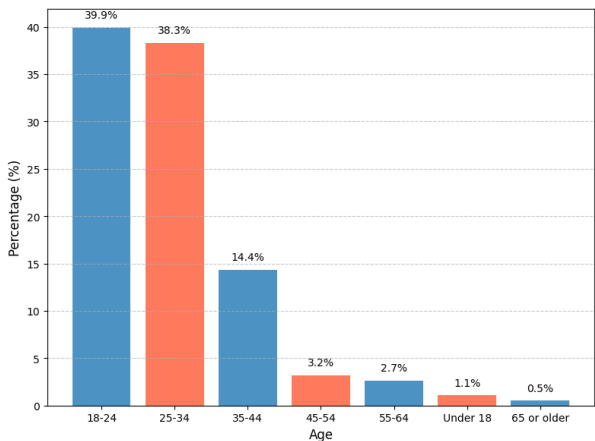


Fig. 2. Age Group Distribution in survey participants

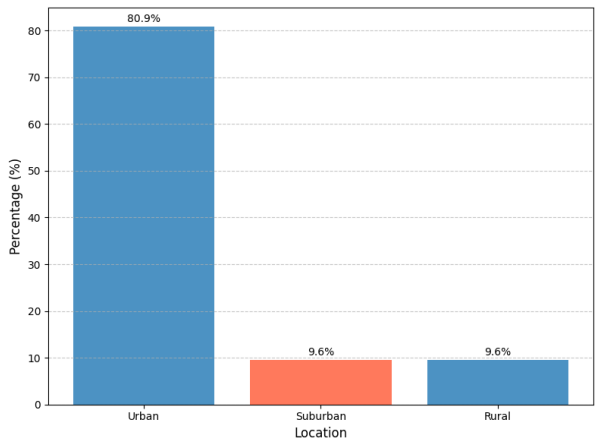


Fig. 3. Location Distribution in survey participants

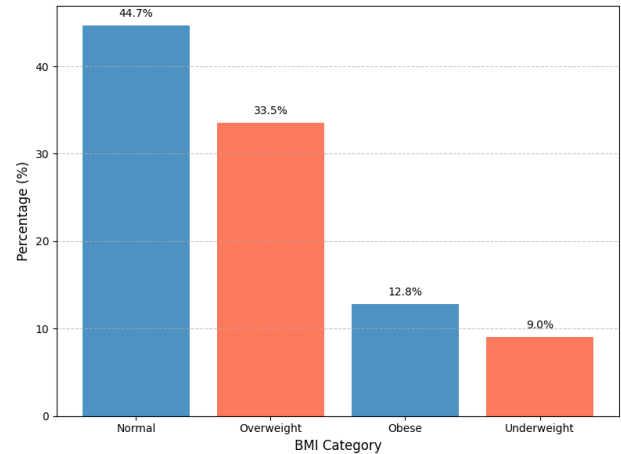


Fig. 4. BMI Category distribution in survey participants

Following table shows summary statistics of EDI of aflatoxin, MOE and Cancer Risk for the participants of the survey:

TABLE II. EDI AND HEALTH RISK INDICATORS SUMMARY STATS FOR SURVEY PARTICIPANTS

Statistics	EDI (ngkg ⁻¹ day ⁻¹)	MOE	Cancer Risk
Mean	45.736070	13.142884	0.921582
Median	42.715683	9.364270	0.860721
Minimum	1.140103	3.045655	0.022973
Maximum	131.334643	350.845484	2.646393

The impact of aflatoxin in terms of Cancer Risk and MOE for each category of categorical variables have been determined in the following tables:

TABLE III. MEAN MOE AND CANCER RISK IN AGE GROUPS

Age Group	MOE	Cancer Risk
18-24	16.356432	0.950862
25-34	10.979543	0.924557
35-44	9.829910	0.955468
45-54	13.846764	0.753489
55-64	12.419561	0.671596
65 or older	15.993240	0.503963
Under 18	13.511767	0.597070

Table III shows that the most impacted age group is 35-44 having highest Cancer Risk and lowest MOE.

TABLE IV. MEAN MOE AND CANCER RISK IN GENDERS

Gender	MOE	Cancer Risk
Male	9.983855	0.987675
Female	20.981957	0.757572

Table IV shows that the most impacted gender is Male having highest Cancer Risk and lowest MOE

TABLE V. MEAN MOE AND CANCER RISK IN LOCATIONS

Location	MOE	Cancer Risk
Rural	10.336261	0.999813
Suburban	12.098756	0.920292
Urban	13.598894	0.912470

Table V shows that the most impacted Location is Rural having highest Cancer Risk and lowest MOE

TABLE VI. MEAN MOE AND CANCER RISK IN BMI CATEGORIES

BMI Category	MOE	Cancer Risk
Underweight	7.092990	1.321843
Normal	10.154620	0.966149
Overweight	17.812057	0.817852
Obese	15.630571	0.754371

Table VI shows that the most impacted BMI Category is Underweight having highest Cancer Risk and lowest MOE

TABLE VII. MEAN MOE AND CANCER RISK POSED BY MILK TYPE

Consumed Milk Type	MOE	Cancer Risk
Fresh Milk	12.747424	0.951972
Packaged Milk	15.402655	0.747922

Table VII shows that the users consuming raw/fresh milk are more impacted having higher Cancer Risk and lower MOE than those consuming packaged milk.

B. Correlation Analysis

Following scatterplots show relation of Wheat EDI, Rice EDI and Milk EDI with MOE and Cancer Risk with correlation coefficients:

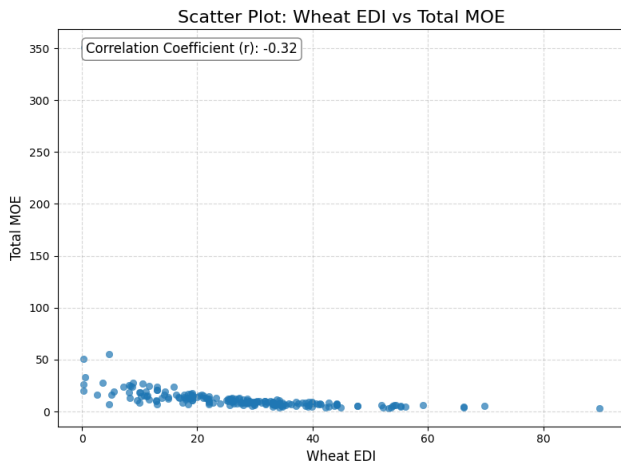


Fig. 5. Scatter Plot: Wheat EDI vs. MOE

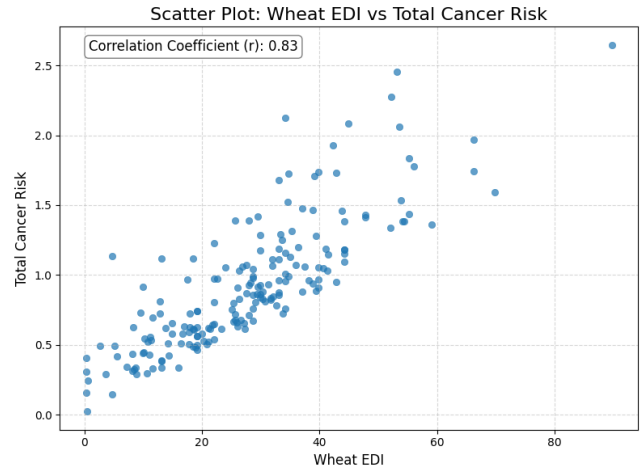


Fig. 6. Scatter Plot: Wheat EDI vs. Cancer Risk

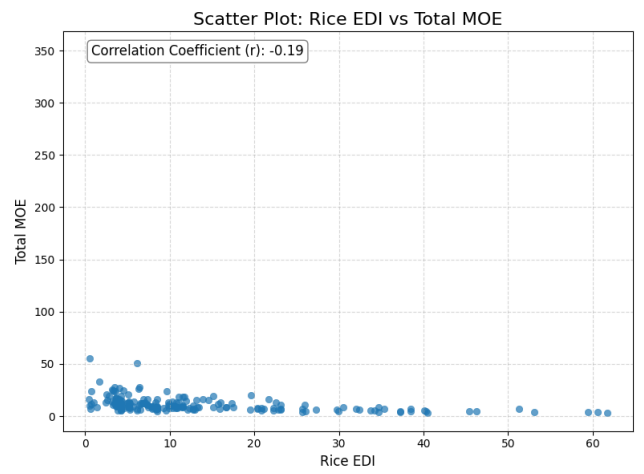


Fig. 7. Scatter Plot: Rice EDI vs. MOE

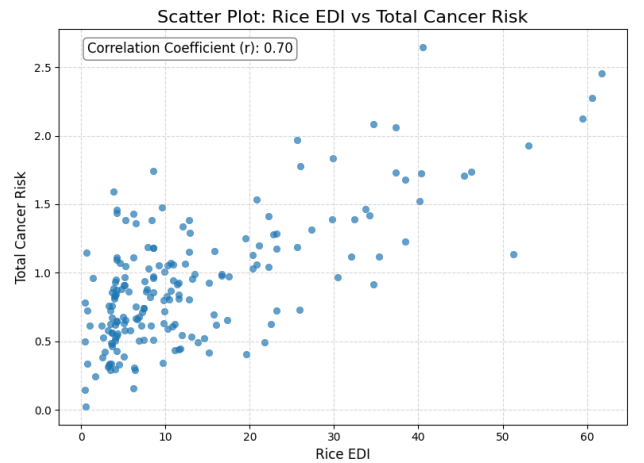


Fig. 8. Scatter Plot: Rice EDI vs. Cancer Risk

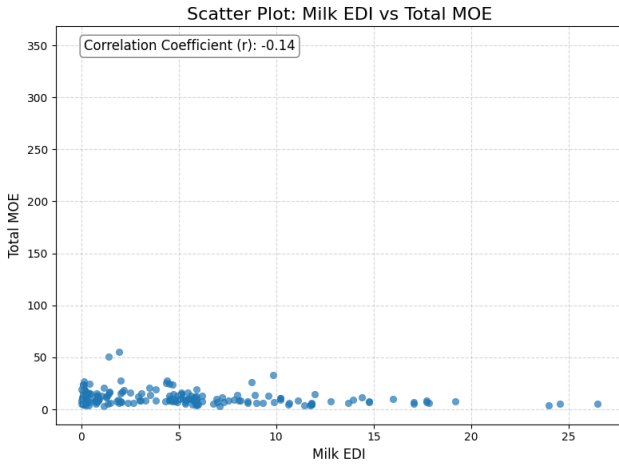


Fig. 9. Scatter Plot: Milk EDI vs. MOE

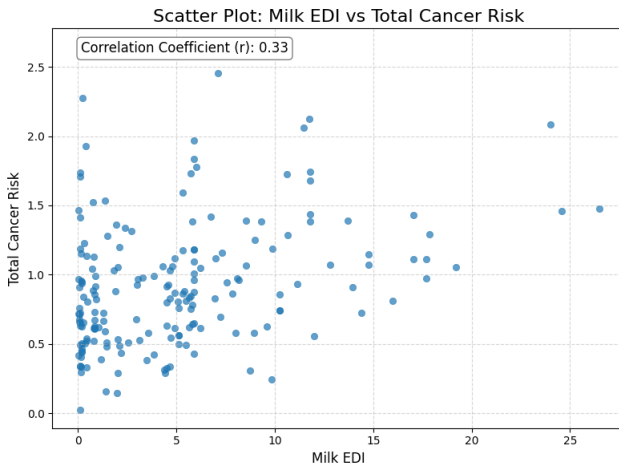


Fig. 10. Scatter Plot: Milk EDI vs. Cancer Risk

Figures 5 to 10 shows that Wheat EDI has the highest correlation with MOE and Cancer Risk depicting that aflatoxins in wheat provide the highest risk for causing cancer. Milk, on the other hand, has the lowest correlation with both MOE and Cancer Risk showing its weak impact on contributing towards cancer.

C. Hypothesis Testing

Following table shows results of T-Test and ANOVA performed for determining whether group mean for MOE and Cancer Risk for various categories are equal or not with:

Null Hypothesis (H_0): All group means are equal

Alternative Hypothesis (H_1): At least one group mean is different.

And null hypothesis is rejected if $p \leq 0.05$

TABLE VIII. T-TEST AND ANOVA TEST RESULTS

Hypothesis Test	Test	p-value	Result
T-Test	Gender vs. MOE	0.0077	Reject H_0
T-Test	Gender vs. Cancer Risk	0.0018	Reject H_0

ANOVA	BMI Category vs. MOE	0.2256	Accept H_0
ANOVA	BMI Category vs. Cancer Risk	0.0001	Reject H_0
ANOVA	Age vs. MOE	0.9094	Accept H_0
ANOVA	Age vs. Cancer Risk	0.6113	Accept H_0
ANOVA	Location vs. MOE	0.8659	Accept H_0
ANOVA	Location vs. Cancer Risk	0.7512	Accept H_0

Thus, Table VIII shows that Cancer Risk and MOE means vary among categories in Gender and BMI. For rest of the categorical variables, Cancer risk and MOE means can be estimated to be same depicting no significant difference of aflatoxin impact among these categories.

D. Confidence Intervals

Lastly, we aim to determine population parameters for EDI, MOE and Cancer Risk. Following plots show distribution of these three parameters:

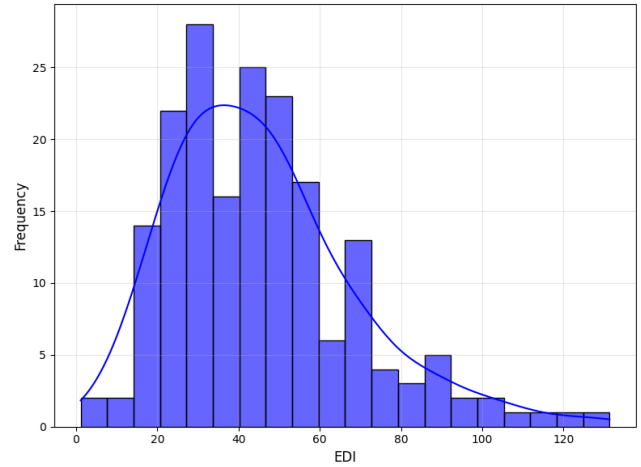


Fig. 11. Distribution of Aflatoxin EDI in survey participants

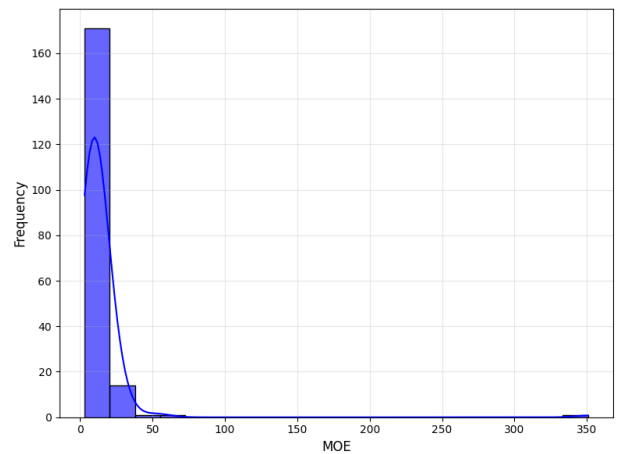


Fig. 12. Distribution of MOE in survey participants

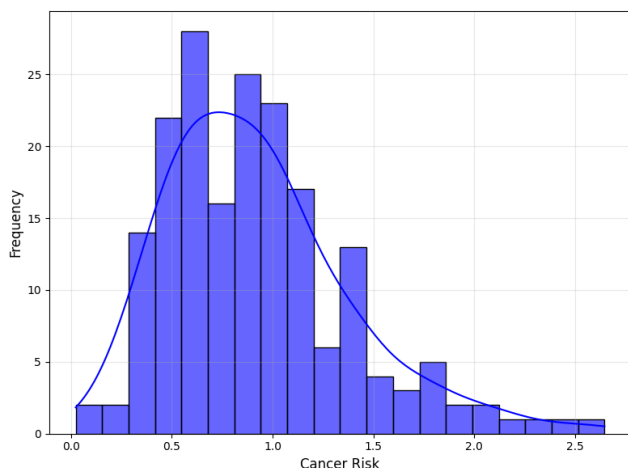


Fig. 13. Distribution of Cancer Risk in survey participants

Figures 11 to 13 shows that EDI and Cancer Risk are approximately normally distributed whereas MOE is highly skewed with most of the values lying in smaller bins. Since, EDI and Cancer Risk can be approximated to be normally distributed and MOE has sample size more than 30 so we can use t-statistic for finding confidence interval of population means for these three parameters. We used 95% confidence interval for this analysis. Results are summarized below:

TABLE IX. 95% CI POPULATION MEAN ESTIMATION FOR AFLATOXIN EDI, MOE AND CANCER RISK

Parameter	95% Confidence Interval
EDI	(42.4432, 49.0289)
MOE	(9.4368, 16.8490)
Cancer Risk	(0.8552, 0.9879)

V. CONCLUSION

Aflatoxin consumption can be as dangerous as exposure to any other toxin causing cancer and unfortunately we are exposed to this toxin every day by consuming the most commonly used diets containing wheat, rice and milk. Our analysis shows that the survey participants on average consume $45.736070 \text{ ngkg}^{-1}\text{day}^{-1}$ of aflatoxin posing a mean Cancer Risk of 0.921582 cancer/year/100,000 population ng/kg bw/day and mean MOE of 13.142884 which is way below the safe limit of 10,000 as stated by European Food Safety Authority. The age group most affected by aflatoxin is found to be 35-44. Among genders, Males are found to be at high risk for cancer due to aflatoxin consumption. Rural regions among the regions under study have the highest aflatoxin consumption. For BMI levels, individuals falling in Underweight category are found to be at high risk. Lastly, it is found that people consuming raw milk are more exposed to aflatoxin than the people consuming packaged milk.

Correlation analysis performed concludes that wheat intake has the overall highest contribution to aflatoxin consumption among other diets studied and thus aflatoxin in wheat poses highest cancer threat.

Hypothesis testing for studying difference in MOE and Cancer Risk means among categories of different groups analyzed shows that MOE and Cancer Risk varies among genders and BMI levels. For regions and age groups, they are

almost same depicting that all regions and age groups are equally affected by aflatoxin consumption.

Lastly, we estimated population parameters for aflatoxin EDI, MOE and Cancer Risk through sample statistics. 95% confidence intervals for EDI, MOE and Cancer Risk are found to be (42.4432, 49.0289), (9.4368, 16.8490) and (0.8552, 0.9879) respectively.

High intake of aflatoxins through diet and above safe limit exposure and cancer risk levels determined in this analysis calls for an urgent need of action from policy makers and implementers. Lack of check and balance on the health of crops is one of the biggest factors contributing towards such high contamination of aflatoxin in food. Another biggest factor towards its widespread is the lack of awareness regarding aflatoxins and the food most contaminated with it among the individuals. Our survey has also shown that 78.7% of people were unaware of the term aflatoxin and that it is present in common diet. Strict regulations and rising awareness regarding aflatoxin among people is the need of hour and therefore, must be done on urgent basis.

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